CH 6 - Eigenvectors and Diagonalization

Luke Lu • 2026-04-01

Matrix of a Linear Mapping, Similar Matrices

Recall the derivation for the standard matrix $[L]$ of a linear mapping $L : \mathbb{R}^n \to \mathbb{R}^n$ for an arbitrary $\vec{x} \in \mathbb{R}^n$, we have

$$L(\vec{x}) = L(x_1\vec{e}_1 + ... + (x_n\vec{e}_n)) = x_1L(\vec{e}_1) + ... + x_nL(\vec{e}_n) = ... = [L]\vec{x}$$

Now we wish to generate an arbitrary basis $B = \{\vec{v}_1, ..., \vec{v}_n\}$ of $\mathbb{R}^n$.

Now the goal is to find a matrix $[L]_B$ s.t.

$$[L(\vec{x})]_B = [L]_B[\vec{x}]_B$$

$$\begin{aligned}
[L(\vec{x})]_B &= [L(b_1\vec{v}_1 + ... + b_n\vec{v}_n)]_B, \quad b_1, ..., b_n \in \mathbb{R} \\
&= [b_1L(\vec{v}_1) + ... + b_nL(\vec{v}_n)]_B \\
&= b_1[L(\vec{v}_1)]_B + ... + b_n[L(\vec{v}_n)]_B \\
&= \left[[L(\vec{v}_1)]_B \ ... \ [L(\vec{v}_n)]_B\right] \begin{bmatrix} b_1 \\ ... \\ b_n \end{bmatrix} \\
&= [L]_B[\vec{x}]_B
\end{aligned}$$

🐳 **Info — B-Matrix**

Let $B = \{\vec{v}_1, ..., \vec{v}_n\}$ be a basis for $\mathbb{R}^n$ and let $L : \mathbb{R}^n \to \mathbb{R}^n$ be a linear operator. The B-matrix of L is defined as

$$[L]_B = \left[[L(\vec{v}_1)]_B \ ... \ [L(\vec{v}_n)]_B\right]$$

It satisfies $[L(\vec{x})]_B = [L]_B[\vec{x}]_B$

Examples:

Let $\vec{a} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}, n = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$

Let $B = \{\vec{a}, \vec{n}\}$ for $\mathbb{R}^n$

Let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be the mapping given by $L(\vec{x}) = \text{proj}_{\vec{a}}(\vec{x})$

1. $[L]$

   $L(\vec{e}_1) = \text{proj}_{\vec{a}}(\vec{e}_1) = \frac{\vec{e}_1 \cdot \vec{a}}{\|\vec{a}\|^2}\vec{a} = \frac{5}{34}\begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{25}{34} \\ \frac{15}{34} \end{bmatrix}$

   $L(\vec{e}_2) = \text{proj}_{\vec{a}}(\vec{e}_2) = \frac{\vec{e}_1 \cdot \vec{a}}{\|\vec{a}\|^2}\vec{a} = \frac{3}{34}\begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{15}{34} \\ \frac{9}{34} \end{bmatrix}$

   $[L] = \begin{bmatrix} \frac{25}{34} & \frac{15}{34} \\ \frac{15}{35} & \frac{9}{34} \end{bmatrix}$

2. $[L]_B$

$$L(\vec{a}) = \vec{a} = \vec{a} + 0\vec{n}$$
$$= [L(\vec{a})]_B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$L(\vec{n}) = \vec{0} = 0\vec{a} + 0\vec{n}$$
$$= [L(\vec{a})]_B = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$[L]_B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

🐳 **Info — Diagonal Matrix**

A $n \times n$ matrix D is said to be a diagonal matrix if $d_{ij} = 0, \forall i \neq j$. Denoted

$$\text{diag}(d_{11}, d_{22}, \ldots, d_{nm})$$

Similar Matrix

Let $A = [L] = [L(\vec{e}_1) \ \ldots \ L(\vec{e}_n)]$

Let $B = [L]_B$

Let $P = {}_SP_B = [\vec{v}_1 \ \ldots \ \vec{v}_n]$ where $S = \{\vec{e}_1 \ \ldots \ \vec{e}_n\}$

Note that $[x]_S = {}_SP_B[\vec{x}]_B \implies \vec{x} = P[\vec{x}]_B \implies [\vec{v}]_B = P^{-1}\vec{x}$

$$B = [L]_B = \left[[L(\vec{v}_1)]_B \ \ldots \ L(\vec{v}_n)\right]_B$$
$$= \left[[[L]\vec{v}_1]_B \ \ldots \ [[L]\vec{v}_n]_B\right]$$
$$= \left[[A\vec{v}_1]_B \ \ldots \ [A\vec{v}_n]_B\right]$$
$$= [P^{-1}(A\vec{v}_1) \ \ldots \ P^{-1}(A\vec{v}_n)]$$
$$= [(P^{-1}A)\vec{v}_1 \ \ldots \ (P^{-1}A)\vec{v}_n]$$
$$= P^{-1}A[\vec{v}_1 \ \ldots \ \vec{v}_n]$$
$$= P^{-1}AP \text{ or } [L]_B = {}_SP_B^{-1}[L]_S{}_SP_B$$

🐳 **Info — Similar Matrices**

If A and B be $n \times n$ matrices s.t. $P^{-1}AP = B$ for some invertible matrix P, t

1. A is said to be **similar** to B
2. rank A = rank B
3. det A = det B
4. tr A = tr B where tr A is defined by tr $A = \sum_{i=1}^{n}(a_{ii})$ and is called the **trace** of a matrix

Note:

If A is similar to B, $\exists P$ invertible s.t.

$$\begin{aligned} B &= P^{-1}AP \\ PB &= PP^{-1}AP \\ PB &= AP \\ A &= PBP^{-1} \end{aligned}$$

Eigenvalues and Eigenvectors

With $A = [L]$ and $P = [\vec{v}_1 ... \vec{v}_n]$

$D = \text{diag}(\lambda_1, \lambda_2, ..., \lambda_n) = \left[\lambda_1\vec{v}_1 \ \lambda_2\vec{2} \ ... \ \lambda_n\vec{v}_n\right]$

We want $D = P^{-1}AP \implies PD = AP$

Setting $AP = PD$, we get

$$\begin{aligned} A[\vec{v}_1 \ ... \ \vec{v}_n] &= P[\lambda_1\vec{e}_1 \ ... \ \lambda_2\vec{e}_n] \\ [A\vec{v}_1 \ ... \ A\vec{v}_n] &= [P(\lambda_1\vec{e}_1) \ ... \ P(\lambda_2\vec{e}_n)] \\ &= [\lambda_1(P\vec{e}_1) \ ... \ \lambda_n(P\vec{e}_n)] \\ &= [\lambda_1\vec{v}_1 \ ... \ \lambda_n\vec{v}_n] \qquad \text{(By Column Extraction Theorem)} \end{aligned}$$

$\implies A\vec{v}_1 = \lambda_i\vec{v}_i, \forall i = 1, 2, ..., n$

Meaning that the effect of multiplying A by $\vec{v}_i$ is the same as scaling $\vec{v}_i$ by some λ_i

🐳 **Info — Eigenvalues, Eigenvectors, Eigenpairs**

1. Let A be $n \times n$ matrix. If there exists a vector $\vec{v} \neq \vec{0}$ s.t. $A\vec{v} = \lambda\vec{v}$, then the scalar λ is the **eigenvalue** of A and $\vec{v}$ is called **eigenvectors** of A corresponding to λ. The pair $(\lambda, \vec{v})$ is called an **eigenpair**
2. Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be a linear operator. If there exists a vector $\vec{v} \neq \vec{0}$ s.t. $L(\vec{v}) = \lambda\vec{v}$, then λ is caled an **eigenvalue** of L and $\vec{v}$ is called an **eigenvector** of L corresponding λ
3. A scalar λ is an eigenvalue of a $n \times n$ matrix A if and only if $C_{A(\lambda)} = 0$

How to find Eigenvalues

We need $\vec{v} \neq \vec{0}$ s.t. $A\vec{v} = \lambda\vec{v}$

$A\vec{v} - \lambda\vec{v} = \vec{0} \implies = (A - \lambda I)\vec{v} = \vec{0}$

To get non-zero $\vec{v}$ solution, $\det(A - \lambda I) = 0$

Examples:

1. Let $A = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}, \vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

   $A\vec{v} = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ = 2 vec(v).

   $\lambda = 2$ is an eigenvalue of A correspoind to the eigenvector $\vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

2. Let $A = \begin{bmatrix} 0 & 1 \\ 3 & -2 \end{bmatrix}$, find eigenvalues and eigenvectors of A

$$0 = \det\,(A - \lambda I) = \det\left(\begin{bmatrix} 0 & 1 \\ 3 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) = \begin{bmatrix} -\lambda & 1 \\ 3 & -2-\lambda \end{bmatrix}$$
$$= -\lambda(-2-\lambda) - 3 = (\lambda+3)(\lambda-1) = 0$$

$\implies \lambda_1 = 1, \lambda_2 = -3$

$A - \lambda_1 I = \begin{bmatrix} -1 & 1 \\ 3 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \implies \vec{v}_1 = t\begin{bmatrix} 1 \\ 1 \end{bmatrix}, t \in \mathbb{R} \setminus \{0\}$

$A - \lambda_2 I = \begin{bmatrix} 3 & 1 \\ 3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1}{3} \\ 0 & 0 \end{bmatrix} \implies \vec{v}_2 = s\begin{bmatrix} -1 \\ 3 \end{bmatrix}, s \in \mathbb{R} \setminus \{0\}$

🐳 **Info — Eigenspace**

Let A be a $n \times n$ matrix with eigenvalue λ. We call the nullspace of $A - \lambda I$ the **eigenspace** of λ, denoted E_λ

Let A be a $n \times n$ matrix with eigenvalue λ_1.

- **Algebraic Multiplicity**
  - The algebraic multiplicity of λ_1, denoted a_{λ_1}, is the number of times that λ_1 is a root of the characteristic polynomial $C(\lambda)$. That is, if $C(\lambda) = (\lambda - \lambda - 1)^k C_1(\lambda)$, where $C_1(\lambda_1) \neq 0$, then$a_{\lambda_1} = k$
- **Geometric Multiplicity**
  - The geometric multiplicity of λ_1, denoted g_{λ_1}, is the dimension of its eigenspace, $g_{\lambda_1} = \dim\left(E_{\lambda_1}\right)$

1. Let A and B be similar matrices, then A and B have the same characteristic polynomial, and hence the same eigenvalue.
2. If A is a $n \times n$ matrix with the eigenvalue λ_1, then $1 \leq g_{\lambda_1} \leq a_{\lambda_1}$

Examples:

Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Find the eigenvalues of A, bases for the corresponding eigenspace, state the geometric and algebraic multiplicity of each eigenvalue.

$$0 = \det(A - \lambda I) = \det\left(\begin{bmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{bmatrix}\right)$$
$$= -\lambda \det\begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} - \det\begin{bmatrix} 1 & 1 \\ 1 & -\lambda \end{bmatrix} + \det\begin{bmatrix} 1 & -\lambda \\ 1 & 1 \end{bmatrix} = -\lambda(\lambda-1)(\lambda+1) + (\lambda+1) + (\lambda+1)$$
$$= -(\lambda+1)(\lambda^2 - \lambda - 2) = -(\lambda+1)^2(\lambda-2) \implies$$

$\lambda_1 = -1, \lambda_2 = 2$

$$A - \lambda_1 I = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ & 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies s\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \implies B_{\lambda_1}\left\{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\right\}.$$

Bases for $\lambda_1 = \text{Span } B_{\lambda_1}$

$$A - \lambda_2 I \implies r\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \implies B_{\lambda_2}\left\{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right\}.$$

Bases for $\lambda_2 = \text{Span } B_{\lambda_2}$

For $\lambda_1 = -1, a_{\lambda_1} = 2, g_{\lambda_1} = 2$

For $\lambda_2 = 2, a_{\lambda_1} = 1, g_{\lambda_1} = 1$

Proof

Eigenspace Theorem 1

Suppose that $\exists P$(invertible) s.t. $B = P^{-1}AP$, then

$$\begin{aligned} B - \lambda I &= P^{-1}AP - \lambda I = P^{-1}AP - \lambda P^{-1}P \\ &= P^{-1}AP - \lambda P^{-1}IP = P^{-1}(AP - \lambda IP) = P^{-1}((A - \lambda I)P) = A - \lambda I \end{aligned}$$

Thus $\det\ B - \lambda I = \det\ A - \lambda I$

□

Diagonalization

🐳 **Info — Diagonalization Theorem**

A $n \times n$ matrix $A \in M_{n\times n}(\mathbb{R})$ is said to be **diagonalizable** if A is similar to a diagonal matrix $D \in M_{n\times n}(\mathbb{R})$. If $P^{-1}AP = D$, then P diagonalizes A.

- A $n \times n$ matrix A is diagonalizable (over $\mathbb{R}$) if and only if there exists a basis $\{\vec{v}_1, ..., \vec{v}_n\}$ for $\mathbb{R}^n$ of eigenvector of A
- If A is a $n \times n$ matrix with eigenpairs $(\lambda_1, \vec{v}_1), ..., (\lambda_k, \vec{v}_k)$ where $\lambda_i \neq \lambda_j$ for $i \neq j$, then $\{\vec{v}_1, ..., \vec{v}_k\}$ is linearly independent.
- If A is a $n \times n$ matrix with distinct eigenvalues $\lambda_1, ..., \lambda_k$ and $B_i = \left\{\vec{v}_{i,1}, ..., \vec{v}_{i,g_{\lambda_i}}\right\}$ is a basis for the eigenspace of λ_i for $1 \leq i \leq k$, then $B_1 \cup B_2 \cup ... \cup B_k$ is a linearly independent set
- If A is a $n \times n$ matrix with n distinct eigenvalues, then A is diagonalizable.

Diagonalizability Test

If A is a $n \times n$ matrix whose characteristic polynomial factors as

$$C_A(\lambda) = (\lambda - \lambda_1)^{a_{\lambda_1}} ... (\lambda - \lambda_k)^{a_{\lambda_k}}$$

where $\lambda_1, ..., \lambda_k$ are the distinct eigenvalues of A, then A is diagonalizable if and only if $g_{\lambda_i} = a_{\lambda_i}$ for $1 \leq i \leq k$.

Lemma

If A is a $n \times n$ matrix with n distinct eigenvalues, then A is diagonalizable.

Theorem

- If $\lambda_1, ..., \lambda_n$ are all the n eigenvalues of a $n \times n$ matrix A(repeated according to algebraic multiplicity), then

$$\det\ A = (\lambda_1)...(\lambda_n) \text{ and } \operatorname{tr}\ A = \lambda_1 + ... + \lambda_n$$

Example:

Let $A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

Diagonalize A if possible, that is, for an invertible matrix P and a diagonal matrix D s.t.

$PAP^{-1} = D$

$C(\lambda) = 0 \implies 0 = \det(A - \lambda I) = \det \begin{bmatrix} 2-\lambda & 2 & 2 \\ 2 & -2-\lambda & 0 \\ 2 & 0 & 1-\lambda \end{bmatrix} = -\lambda(\lambda - 4)(\lambda + 3)$

$\implies \lambda_1 = 0, \lambda_2 = 4, \lambda_3 = -3$ are eignvalues of A

Since A has 3 distinct eigenvalues, thus it is diagonalizable

$\lambda_1 = 0 \implies \begin{bmatrix} 2 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \implies \vec{v} = s \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, s \in \mathbb{R}$

$B_1 = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$ is a basis for E_{λ_1}

$\lambda_2 = 4 \implies \begin{bmatrix} -2 & 2 & 2 \\ 2 & -6 & 0 \\ 2 & 0 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{3}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \implies \vec{v} = t \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}, t \in \mathbb{R}$

$B_2 = \left\{ \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \right\}$ is a basis for E_{λ_2}

$\lambda_3 = -3 \implies \begin{bmatrix} 5 & 2 & 2 \\ 2 & 1 & 0 \\ 2 & 0 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \end{bmatrix} \implies \vec{v} = q \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}, q \in \mathbb{R}$

$B_3 = \left\{ \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix} \right\}$ is a basis for E_{λ_3}

Let $P = [\vec{v}_1 \ \vec{v}_2 \ \vec{v}_3] = \begin{bmatrix} 1 & 3 & -2 \\ 1 & 1 & 4 \\ -2 & 2 & 1 \end{bmatrix}, D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -3 \end{bmatrix}$

Powers of Matrices

🐳 **Info — Powers of Matrices**
Let A be a $n \times n$ matrix. If there exists a matri P and diagonal matrix D s.t. $P^{-1}AP = D$, then

$$A^k = PD^kP^{-1}$$

Proof

$D = P^{-1}AP \implies A = PDP^{-1} \implies$

$A^k = (PDP^{-1})^k = (PDP^{-1})(PDP^{-1})(PDP^{-1})...(PDP^{-1}) = PD(P^{-1}P)D(P^{-1}P)...(P^{-1}P)DP^{-1}$

$= PD^kP^{-1}$

□

Example:

Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. Diagonalize A and compute A^{307}

It can be show that $\lambda_1 = -1, \lambda_2 = 3$ are eigenvalues of A with corresponding eigenvectors $\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ respectively.

Let $P = [\vec{v}_1 \ \vec{v}_2] = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$

Then $D = P^{-1}AP, P^{-1} = \frac{1}{2}\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}, D^{307} = \begin{bmatrix} -1 & 0 \\ 0 & 3^{307} \end{bmatrix}$

$A^{307} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 3^{307} \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -1+3^{307} & 1+3^{306} \\ 1+3^{307} & -1+3^{307} \end{bmatrix}$

Complex Numbers

Definitions for Complex Numbers

$$\mathbb{C}^n = \left\{ \begin{bmatrix} z_1 \\ z_2 \\ \dots \\ z_n \end{bmatrix} \mid z_1, z_2, \dots, z_n \in \mathbb{C} \right\}$$

Additional and scalar multiplication in the same ways as $\mathbb{R}^n$

$M_{m\times n}(\mathbb{C})$ is the set of all $m \times n$ matrices with complex entries.

For $A \in M_{m\times n}(\mathbb{C})$, $\overline{A}$ is defined by $\left(\overline{A}_{ij}\right) = \overline{a}_{i_j}$

For $\vec{z} \in \mathbb{C}^n$, $\overline{\vec{z}} = \begin{bmatrix} \overline{z_1} \\ \dots \\ \overline{z_n} \end{bmatrix}$

$A = [\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n]$, $\overline{A\vec{z}} = \overline{z_1\vec{a}_1 + \dots + z_n\vec{a}_n} = \overline{z_1\vec{a}_1} + \dots + \overline{z_n}\vec{a}_n = \overline{z_1}\overline{\vec{a}_1} + \dots + \overline{z_n}\overline{\vec{a}_n}$

$$= \left[\overline{\vec{a}_1} \ \dots \ \overline{\vec{a}_n}\right] \begin{bmatrix} \overline{z_1} \\ \dots \\ \overline{z_n} \end{bmatrix}$$

Examples:

1. Let $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, diagonalize A over $\mathbb{C}$

   0 $\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & -1 \\ 1 & 1-\lambda \end{bmatrix} = (1-\lambda)^2 + 1$

   $-1 = (1-\lambda)^2 \implies 1 - \lambda = \pm i \implies \lambda = 1 \pm i$

   $A - \lambda_1 I = \begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} (-i)R_1 \sim \begin{bmatrix} 1 & i \\ 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \implies x_1 = -ix_2 = -it \implies \vec{x} = t\begin{bmatrix} -i \\ 1 \end{bmatrix}, t \in \mathbb{C}$

   $B_1 = \left\{ \begin{bmatrix} -i \\ 1 \end{bmatrix} \right\}$ is a basis for E_{λ_1}

   Similarily, $A - \lambda_2 I$ have the same process.

   $B_2 = \left\{ \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$ is a basis for E_{λ_2}

   Let $P = \begin{bmatrix} -i & i \\ -1 & 1 \end{bmatrix}, D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} 1-i & 0 \\ 0 & 1+i \end{bmatrix}$

   These satisfy $D = P^{-1}AP$

2. Compute A^{100} from (1)

   $A^{100} = PD^{100}P^{-1}$

   $P^{-1} = \frac{1}{-2i} \begin{bmatrix} 1 & -i \\ -1 & -i \end{bmatrix}$

   $\overset{\frac{1}{i}=-i}{=} \frac{1}{2} \begin{bmatrix} i & 1 \\ -i & 1 \end{bmatrix}$

   $D^{100} = \begin{bmatrix} \sqrt{2}(\cos 25\pi - i \sin 25\pi) & 0 \\ 0 & \sqrt{2}(\cos 25\pi + \sin 25\pi) \end{bmatrix} = \begin{bmatrix} -2^{50} & 0 \\ 0 & -2^{50} \end{bmatrix}$

   $A^{100} = \begin{bmatrix} -i & i \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -2^{50} & 0 \\ 0 & -2^{50} \end{bmatrix} \frac{1}{2} \begin{bmatrix} i & 1 \\ -i & 1 \end{bmatrix} = \begin{bmatrix} -2^{50} & 0 \\ 0 & -2^{50} \end{bmatrix}$

🐳 **Info — Theorem**

Let $A \in M_{n \times n}(\mathbb{R})$. If λ is a non-real eigenvalue of A with corresponding eigenvector $\vec{z}$, then $\overline{\lambda}$ is also an eigenvalue of A with eigenvector $\overline{\vec{z}}$

Also, if n is odd, A has at least one real eigenvalue.

END OF FINAL